Noldus
Information Technology

CatWalk XT 10.6

Technical Specifications

CatWalk™ XT is an easy to use and highly sensitive tool for the detailed assessment of gait changes in rats and mice. It is a plug and play system and consists of the following components:

- The CatWalk XT walkway, consisting of a hardened glass floor plate, a spring operated corridor, illuminated ceiling, control unit, and goal box.
- A 100 frames per second (fps) high speed color camera that records the footfalls and body contour. (See Part I – Hardware)
- The CatWalk XT software, version 10 for the recording and analysis of the footprints and gait of rodents traversing the walkway. (See Part II – Software)
- Top-of-the-line computer equipment for running the software.

PART 1 - HARDWARE

WALKWAY

The CatWalk XT walkway consists of a scratch resistant glass plate that is traversed by a rodent from left to right, towards the goal box . It facilitates the recording of footprints and limits the movement of an animal to elicit uniform runs. Although the system works best in a low light environment, a dim red light source can be used to illuminate the working environment. The dimensions of the walkway are 130 x 68 x 152 cm (51.2" x 26.8" x 59.8").

ILLUMINATED FOOTPRINTS TECHNOLOGY

TThe Illuminated Footprints technology is based on internal light reflection; green LED light is injected into the glass plate and is completely internally reflected, except for those areas where the animal makes contact with the glass plate (typically the animal's paws) light escapes and is reflected by the paw towards the camera. The high speed color camera (100 fps) that is positioned underneath the glass plate captures the now illumnated footprints and their position in space and time is determined by the CatWalk XT software.

The Illuminated Footprints™ technology assures that the actual footprints of the rodent can be captured with a high speed video camera while the paws touch the walkway. This ensures maximal spatial and temporal resolution and it is why CatWalk XT is able to produce such high-quality data compared to other gait analysis systems. This technology is sensitive enough to identify intensity differences between paws as a result of how the animal distributes its body weight. The captured video is processed by the CatWalk XT software. Based on the timing, dimensions, position, dynamics, and intensity of each footfall, numerous parameters are calculated for qualitative and quantitative analysis of footfalls and gait.

GLASS PLATE

The glass plate of the walkway is made of clear, hardened, scratch resistant, safety glass with polished edges. This type of glass is less contaminated than standard glass in order to assure evenly distributed lighting conditions along the length and width of the walkway. A replacement glass plate can be ordered from the Noldus website when necessary. Dimensions of the glass plate are 130 x 20 x 0.5 cm (51.2" x 7.9" x 0.2").

SPRING OPERATED CORRIDOR

The corridor consists of two black walls to limit and direct an animal's movement over the glass plate. The width of the corridor can be easily adjusted in a continuous manner making the system suitable for both mice and rats of all sizes. On the left side the corridor walls are bended outwards so animals can be easily placed on a closed system. On the top of the corridor walls an extra diffusion plate is mounted. Besides more even lighting this will give stability to the walls and will prevent animals from peeking over the corridor. If necessary the diffusion plate can be removed to allow, for instance, tethered animals to traverse the walkway.

For cleaning purposes and in order to take a rodent from the glass plate, the corridor can be effortlessly lifted with just two fingers due to the spring operated lifting mechanism situated at the back of the corridor.

GOAL BOX

The goal box is located at the right end of the corridor. It consists of a platform with a hole in the middle. On this platform a light-weight box is positioned that covers the hole and has an opening at the side of the corridor in order to allow entry to the box from the corridor. This box can easily be lifted with one hand for access to an animal inside. Once inside the box the animal has the option to jump through the hole into its home cage located below. The goal box design allows for most of the commonly used Macrolon cages to fit underneath. For a precise fit various inserts are provided.

ILLUMINATED CEILING

The illuminated ceiling on top of the walkway creates a red background so that the contour of the animal can be visualized. This contour is used to determine the position of the illuminated paw relative to the body contour and assign its identity. Furthermore, it is used to start and stop the recording and determine the speed of movement.

POWER SUPPLY

The 24V power supply unit can be operated all over the world using mains voltage from 100 to 240V AC and 50 to 60 Hz and three exchangeable main plug adapters.

CATWALK XT CONTROL UNIT

The CatWalk XT Control Unit takes power from the power supply and distributes it over the camera, green LED's that illuminate the walkway and the red LED's of the Illuminated ceiling. The CatWalk XT control unit is connected to the acquisition PC via a USB (type A to type B) cable, the power going to the green and red LED's (in other words, the green and red LED light intensity) can be controlled from within the software.

CAMERA AND MOUNT

The high speed color camera (100 fps) allows you to capture approximately 30 seconds of video per run. The time limit for capturing a run is determined by numerous factors (such as noise level and image resolution). The camera is connected via a CAT-6 UTP cable to a network card inside the computer running the CatWalk XT software. To get a smaller or bigger portion of the glass plate in view of the camera, it can be moved up or down on a vertical slider. Having approximately 60 cm of the glass plate in view with the video camera results in capturing approximately five to six strides of gait in rats. Decreasing the distance between camera and glass plate in order to have approximately 35 cm in view allows you to acquire the same number of strides for a mouse while increasing the image resolution and light sensitivity.

PART 2 - SOFTWARE

NAVIGATION

Start up

When you start the CatWalk XT software, a task dialogue will appear that will guide you through the application. The dialogue will allow you to easily navigate to relevant parts of the software that are often accessed when starting up CatWalk XT.

Task dialogue

Further task dialogue windows will guide you through the application and offers functions depending on the position in the workflow.

Menu bar

The menu bar located in the top of the user interface allows you to navigate anywhere in the application, independent of the task dialogues.

Experiment explorer

The left pane of the user interface displays the experiment explorer that contains the workflow from top to bottom. Double clicking allows you to quickly jump to different positions in the workflow. It also shows you the acquired trials and runs and their status (selected by the data segmentation profile, classification and whether a run is compliant with the run criteria).

Quick links

The bottom of the user interface offers links to the previous and subsequent phase in the workflow.

SETUP

Define and calibrate the walkway

In this part of the software the camera view needs to be cropped to omit all redundant parts of the camera image. The image needs to be calibrated to match real life dimensions.

Detection settings

The detection settings allow you to define the optimal intensity range for footprint detection and for filtering out background noise. The detection settings dialogue has an auto detection function that will give you a suggestion for the optimal detection settings from where you can optimize the detection of footprints by adjusting the power output to the green LED's. Adjusting the red illuminated ceiling light intensity allows for contour detection.

Experiment settings

In the experiment settings you can define experiment groups and time points which later can be used as independent variables in data selection and analysis. In addition, run criteria can be defined in order to set the thresholds for run duration and speed variation. Note that these criteria become read only for the whole experiment once the first run is acquired. Runs that comply with these criteria are labeled accordingly. However, runs that do not comply with these criteria are not discarded so you can still use them for data segmentation (see data segmentation profiles). The minimum number of runs that need to comply with these criteria can also be defined.

On average, three compliant runs per animal per time point are necessary when using five or six step cycles per run. When using less step cycles per run due to time limitations or motivational issues more than three compliant runs are advised to ensure enough data is acquired to allow statistical significance.

Trial list
The trial list allows you to automatically build a list of trials that you want to acquire for each time point and treatment group. A trial is a unique identifier that contains the treatment group, time points, animal ID and a number of runs. A trial is finished when the number compliant runs set in the run criteria are acquired.

Acquisition settings
The acquisition settings dialogue allows you to set several standard operations regarding runs, for instance: it lets you determine after what duration runs should be automatically aborted. The default is set at 10 seconds.

ACQUISITION

Trial selection
Prior to acquisition, the trial that matches the animal to be tested, must be selected. Alternatively, a new trial can be created and defined. Note: In the experiment explorer you can see how many runs are performed in each trial and how many of them are labeled compliant.

Run acquisition
Before starting a run you are advised to refresh the background for optimal detection fidelity. After the animal is placed on the walkway, its image is captured automatically during the entire time it is in view until it either leaves the view of the camera, or the maximum run duration is exceeded. Feedback is provided after each run regarding the compliance of the run with regards to the defined run criteria.

CLASSIFICATION

Trial & Run selection
You can select the trials and runs that require classification. Run duration, average speed and compliancy are given for reference.

Manual footprint classification
The software automatically identifies the parts of the animal that have made contact with the glass plate, which are typically the animal's footprints. Next, the footprints need to be labeled manually by means of a click and select function. Once a footprint is manually labeled the software automatically jumps to the next unlabeled print. Recognition of the footprints is facilitated by dragging the video slider back and forth to study the dynamics.

The Automatic Footprint Classification Module
With Automatic Footprint Classification™, the classification of each print is done automatically by the built-in classification algorithm. Atypical prints are automatically highlighted for manual classification. Once labeling is completed, both qualitative data and quantitative data are automatically generated by the software.

Adjusting the intensity threshold
In the Adjust Intensity Threshold dialogue you can change the intensity threshold that you have set in the detection settings. This change will only apply to the selected run where the set value in the detection settings will be applied to all to be acquired runs. Changing the intensity threshold might be necessary for individual runs when you have a large variation in background noise between runs, trials, or even time points.

Zooming into the acquired video
By zooming into the run video (up to four times) even the smallest detail of a footfall can be visualized. While zoomed into the video the animal will stay fixed into view which is very convenient for inspection of the run. Manual classification inside the zoomed image is also possible.

SINGLE FOOTPRINT MEASUREMENTS

Interactive Footprints Measurements Module

With Interactive Footprints Measurements™ the angle between any paw and the animals body-axis or alternatively its movement vector, can be measured. Furthermore, the distance between the first and the fifth digit (the toe spread), the second and fourth digit (the intermediate toe spread) and the print length for each paw print can be derived. After indicating the treated side of the tested animal, the Sciatic Functional Index (SFI), Tibial Functional Index, and Peroneal Functional Index will be automatically calculated based on the data derived from the hind paws. These indexes can be used to evaluate the functioning of the Sciatic nerve and its branches over time. You can determine these indexes for both rats and mice.

PROFILE EXPLORER

The profile explorer contains the Data Segmentation Profiles and Parameter Profiles. You can define multiple profiles.

Data Segmentation Profiles

Using a Data Segmentation Profile, you can automatically select only parts from each classified run that comply with three parameters that you set (the minimum number of consecutive steps with a specific speed range and maximum allowed speed variation). This allows you to only select step cycles that are within a certain variation bandwidth, reducing the variation that is introduced by bad runs. It also allows you to include partial data from runs that were not compliant with the run criteria. The amount of data incorporated and discarded for each time point and treatment group is visualized and displayed in the data segmentation dialogue. You can select either the steps, animals or runs to be visualized in order to ensure an even representation.

Parameter Profiles

You can select from the complete list of parameters that will be displayed in the Run Statistics, parameter graphs and will be exported to an Excel file.

VISUALIZATION

Print view

The print view displays the footprints in a way that is analogous to footprint analysis with the inking method.

Sub prints view

The sub prints view displays the accumulation of a footprint on frame basis.

Gait diagram

The gait diagram displays time-based information regarding stances and swings of labeled footprints and other body parts.

Print intensities

The print intensities correlate with the weight exerted during a stance. The colored area displays the mean intensity during a stance for each paw. The red line displays the maximum intensity reached anywhere in the print (in other words, it corresponds with the pixel that shows maximum intensity).

Footfall patterns

The footfall patterns window displays the order in which the paws are placed. Blue filled circles denote the start of a step cycle and the corresponding category to which the pattern belongs is displayed at the bottom (in accordance with Cheng et al. , 1997). Crossed circles do not fit in a regular pattern.

3D Footprints

Individual footprints can be displayed in 3D, with the x- and y-axis corresponding with the footprint dimensions and the z-axis corresponding to intensity.

Parameter Charts

From all data that is selected by the Data Segmentation Profile and Parameter Profile, charts can be automatically generated. These charts are displayed per paw, either per time point or treatment group. There are two types of charts that can be displayed; bar charts or five number summery charts.

ANALYSIS

Parameters

A large number of parameters are calculated. A number of these are based on individual footprints:

- *Initial contact* – Indicates the moment that the stance phase starts. It is expressed in seconds.
- *Max contact* – Indicates the moment during a stance phase for which the footprint is largest. It is expressed in seconds.
- *Max area* – Indicates the size of the footprint at Max contact. It is expressed in square pixels.
- *Intensity* – Indicates the average intensity of Max area. The more weight is exerted, the larger the total area of skin-floor contact per pixel, and thus the brighter the pixel.
- *X* – Indicates the position in horizontal direction of the mass-midpoint at Max contact.
- *Y* – Indicates the position in vertical direction of the mass-midpoint at Max contact.

- *Box width* – Indicates the width of the complete footprint (not only at Max contact, but for all frames that make up a stance). It is expressed in pixels.
- *Box length* – Indicates the length of the complete footprint (not only at Max contact, but for all frames that make up a stance). It is expressed in pixels.
- *Print area* – Indicates the area of the complete print (not only at Max contact, but for all frames that make up a stance). It is expressed in square pixels.
- *Stand* – Indicates the duration of the stance phase. It is expressed in seconds.
- *Swing* – Indicates the duration of the swing phase. It is expressed in seconds.
- *Stride length* – Indicates the distance between successive placements of the same paw. It is expressed in pixels.
- *Max contact at* – Indicates the moment of Max contact in relation to the duration of stance. It is expressed in percentage.
- *Swing speed* – Indicates the speed of a swing. It is computed from stride length and swing duration. It is expressed in pixels/seconds.
- *Stand index* – Indicates the speed with which the paw loses contact with the glass plate. A number of parameters are based on the relative positions between different paws.
- *Base of support* – Indicates the base of support for a particular run. Calculation is based on the average Y-coordinate of the left paw minus the average Y-coordinate of the right paw. This results in the average distance between the two fore paws, and the average distance between the two hind paws. It is expressed in pixels.
- *Print positions* – Indicates the position of the hind paw relative to the previous position of the fore paw. Calculation is based on the X-coordinate of the fore paw minus the X-coordinate of the hind paw. This results in the distance between the two lateral right paws, and the two lateral left paws. Positive values indicate that the hind paw was placed behind the forepaw, and negative values indicate that it was placed before the forepaw.
It is expressed in pixels.
- *Toe Spread* – Is the distance (in Distance Units) between the center of the first and fifth toe of a hind paw.
- *Intermediate Toe Spread* – Is the distance (in Distance Units) between the center of the second and fourth toe of a hind paw.
- *Manual Print Length* – Is the distance (in Distance Units) between the center of the third toe and the heel of a paw.
- *Paw Angle Body Axis* – Is the smallest angle (in degrees) between the Manual Print Length line and the line representing the orientation of the body axis
- *Paw Angle Movement Vector* – Is the smallest angle (in degrees) between the Manual Print Length line and the line representing the direction of movement of the animal's body
- *Sciatic Functional Index* – The Sciatic Functional Index (SFI) is a measure for the functional recovery of the sciatic nerve which innervates the hind paws. Therefore, the SFI is only relevant for hind paws. The SFI is calculated differently for rats and mice.
- *Tibial Functional Index* – The Tibial Functional Index (TFI) is a measure for the functional recovery of the tibial nerve which innervates the hind paws. Therefore, the TFI is only relevant for hind paws. The TFI is calculated differently for rats and mice.
- *Peroneal Functional Index* – The Peroneal Functional Index (PFI) is a measure for the functional recovery of the peroneal nerve which innervates the hind paws. Therefore, the PFI is only relevant for hind paws. The PFI is calculated differently for rats and mice.

A number of parameters are based on time-based relationships between different paws. These primarily indicate interlimb coordination:

- *Cadence* – The number of steps per second.
- *Step sequence* – Indicates the order in which the four paws are placed. Rats and mice display 6 types of regular step sequences. CatWalk XT is able to detect which regular step sequence is displayed by an animal. It is expressed in frequency.
- *Regularity index* – Indicates the degree in which the steps taken fall into one of the regular step sequences. If all steps fall in one of these categories the index is 100%.

- *Phase dispersions* – Indicates the moment of initial contact of a paw, relative to the stride cycle of a reference paw. It is expressed in percentage (see Kloos et al., 2005).
- *Couplings* – Indicates the same as phase dispersions. However, the paw of interest (i.e. the target) can never precede the reference paw (i.e. the anchor).
- *Support formulas* – Indicates the number of paws that are simultaneously on the ground during a stride cycle. It is expressed in absolute frequency.
- *Footfall formulas* – Indicates which paws are simultaneously on the ground during a stride cycle.
- *Standing on* – Indicates the relative duration of contact on no, one, two (diagonal, lateral, girdle), three, or four paws. It is expressed in percentage.
- *Initial dual stance* – The duration (in seconds) of ground contact for both hind paws simultaneously (Coulthard et al., 2002, 2003). Dual Stance is used for gait analysis in pain models.
- *Terminal Dual Stance* – Indicates the second step in a step cycle of a hind paw that the contralateral hind paw also makes contact with the glass plate.

Export

Numerical data can be exported in four ways:

- *On frame basis* – The timestamp, X and Y coordinates, print length, width and area, minimum, mean and maximum intensity and width for each stance and for each frame can be exported to Microsoft 2003 or Microsoft 2007.
- *On run basis* – All parameters and statistics calculated by CatWalk XT on a run basis can be exported to Microsoft 2003 or Microsoft 2007. All runs belonging to the same experiment can be exported to one Excel sheet, with the option to exclude runs that did not comply with the run criteria or that are only partially classified.
- *On trial basis* – The same parameters and statistics can be exported on a trial basis. This option is the average of the runs that belong to the trial. Importantly; it is not the average of the run averages. Instead, all the runs that belong to the trial are treated is a single run which is the basis for trial statistics.
- *Video exports* – In the visualization screen you can export the run video and visualizations to an *.avi file. You can set the playback speed at which you want this video to be played after export.
- *Parameter Charts exports* – In the Parameter Charts screen you can export either each individual chart by right clicking on it or all charts of all selected parameters at once by the 'Export Parameter Charts' function in the file menu. All the charts will be conveniently exported to a PDF file.

PART 3 - REQUIREMENTS

ENVIRONMENT

CatWalk XT must be operated in a low light or red ambient light conditions.

OPERATING SYSTEM

Windows 10 64bit.

COMPUTER

CatWalk XT 10.6 has been tested on:

- A Dell Precision T5820, with Intel Xeon W-2102 2,9 GHz with 8 GB RAM, under Windows 10 - 64bit.
- A Dell Precision T5810, with a Xeon E5-1620 v2 Processor (3.7 GHz) with 8 GB RAM, under Windows 8.1 - 64bit.
- Hard disk space: 250 GB.

MAINTENANCE

Between runs, the walkway needs to be cleaned in order to get rid of hairs, faeces, and urine. Besides the electronics related to the LED sources, the foam stripe that is placed between the glass plate and the light encasement is fragile, and needs to be cleaned carefully. The foam stripe is PVC based and resistant to the following solvents:

- Mineral oils
- Acids
- Alkalis
- Vapors
- Alcohol
- Water

The black rubber stripe attached to the long edge of the glass plate can easily be removed for more thorough cleaning.

PART 4 - SERVICE AND SUPPORT

REFERENCE MANUAL

CatWalk XT comes with a reference manual in which all software functions are described in detail. The manual is also included on the installation CD as PDF file. It can be accessed anywhere in the program by pressing F1.

INSTALLATION AND TRAINING

Each CatWalk system comes with on-site installation of the complete system and a comprehensive training course by a qualified consultant. The training, including course materials for the participants, covers the basic operation of the hardware and software.

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TECHNICAL SUPPORT

With NoldusCare, our highly qualified Support staff is available worldwide via telephone, Skype, email or a team viewer meeting. You can expect an answer within 24 hours. As a registered user of CatWalk XT, you receive a free one-year subscription to NoldusCare.

CONSULTING SERVICES

Our customer support goes far beyond solving technical problems. Our staff includes several consultants with an academic degree in the behavioral sciences, as well as qualified engineers. They are happy to help you with research problems related to the use of CatWalk XT.

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For an exhaustive list visit: https://scholar.google.nl/scholar?hl=nl&as_sdt=0%2C5&q="noldus"+"catwalk"&btnG

CONTACT INFORMATION

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